

Samuele Arabia

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PROFESSIONAL SUMMARY

Computer Engineering student focused on embedded software, real-time systems and hardware/software integration, with hands-on experience developing C++ firmware for avionics applications within the Sapienza Rocket Team. Experience with Raspberry Pi RP2350, FreeRTOS, CMake, object-oriented programming and serial debugging. Complementary experience in full-stack development, microservices, automation and AI-based software systems, with a long-term interest in robotics and systems/software architecture.

EDUCATION

Sapienza University of Rome
B.Sc. in Computer and Control Engineering

Rome, Italy
Sep. 2022 – Expected Dec. 2026

EXPERIENCE

Sapienza Rocket Team
Avionics Software Developer

Rome, Italy
Oct. 2025 – Present

- Developed embedded C++ firmware for real-time flight data acquisition and sensor integration within the avionics subsystem.
- Worked with the Raspberry Pi RP2350 and FreeRTOS to develop software components designed for real-time execution.
- Applied object-oriented programming and CMake-based project organization to structure and maintain embedded software components.
- Performed hardware/software integration, testing and debugging through serial communication and runtime logging.
- Collaborated within a cross-functional engineering team to design, test and troubleshoot avionics software under hardware and resource constraints.

PROJECTS

AEHub – Digital Hub | Next.js, React, Python, Playwright, Docker

Jun. 2026 – Present

- Developing a full-stack platform for centralizing academic data and multimedia management through a unified dashboard.
- Designed a decoupled client-server architecture with a Next.js frontend, Python backend, REST APIs and orchestration services.
- Implemented browser automation and persistent session management using Playwright, with Docker-based deployment and a roadmap toward AI-assisted task orchestration.

CodeForge – Portfolio Platform | Astro, TypeScript, Node.js, Fastify, MongoDB, Docker

May 2026 – Present

- Developed a full-stack platform using a decoupled frontend/backend architecture with Astro, Node.js, Fastify and MongoDB.
- Applied a layered Controller–Service–Repository architecture to improve backend organization, maintainability and separation of concerns.
- Applied production-oriented practices including Docker containerization, automated testing, structured logging and API security.

NT-To-SQL | Python, FastAPI, MariaDB, Ollama, Docker

Jun. 2025 – Aug. 2025

- Developed a full-stack application that translates natural-language questions into executable SQL queries using a locally hosted LLM.
- Designed a containerized microservices architecture separating frontend, backend, database, LLM inference and automated testing services.
- Implemented REST APIs for natural-language search, SQL execution, schema inspection and database operations.

TECHNICAL SKILLS

Programming:	C, C++, Python, Java, JavaScript, TypeScript, SQL.
Embedded & Systems:	Embedded C++, Firmware Development, FreeRTOS, Real-Time Systems, Sensor Integration, Hardware/Software Integration, Object-Oriented Programming.
Hardware:	Raspberry Pi RP2350, Serial Communication.
Software Engineering:	Data Structures, Algorithms, System-Level Architecture, Microservices, REST APIs, CMake.
Web & Backend:	React, Next.js, Astro, Node.js, Fastify, FastAPI, MongoDB, MariaDB.
AI & Automation:	LLM-based Applications, Ollama, Browser Automation, Playwright.
Tools:	Git, Docker, Linux, CLion, Arduino IDE, Picotool.

LANGUAGES

Italian: Native	English: C1+	French: C1+
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INTERESTS

Technology: Embedded Systems, Robotics, Unreal Engine, Game Technology, Quantum Computing	
Sports: Football, Running	Music: Music Production, Soundtrack Production, Songwriting, Rap, Singing
Cinema: Science Fiction, Fantasy	Personal Growth: Mindset, Continuous Learning, Self-Improvement, Neuro-Psychology